ENG3014 _ Lab2: Spectrum Analysis of Waveforms

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Introduction

This experiment investigates how signals appear in the frequency domain, and how modulation changes the distribution of energy across frequencies. Instead of studying signals only in the time domain, the oscilloscope FFT mode and Python analysis allow us to observe the spectra of:

1. A pure sinusoidal signal
2. A 50% duty cycle square wave
3. An Amplitude Modulated (AM)
4. A Frequency Modulated (FM)

This links directly to the Fourier Transform and modulation theory, where communication signals are represented and analysed in the frequency domain. The lab demonstrates physically what the lectures explain mathematically: a signal is a sum of frequency components, and modulation redistributes these components around a carrier.

1.Using the Oscilloscope as a Spectrum Analyzer

A 1 kHz sine wave (200 mV peak to peak) was generated and its FFT was recorded both on the oscilloscope and using Python.

Observation: The FFT shows one dominant peak at approximately 1 kHz, which corresponds to the carrier frequency. The rest of the spectrum contains only low level noise (below $-50\,\text{dBV}$).

Relation to Course Content: A pure sine wave contains only one frequency component, which agrees with the theoretical Fourier Transform:

$$x(t) = A\sin(2\pi f_0 t) \quad \Longrightarrow \quad X(f) = \delta(f - f_0)$$

This verifies correct FFT operation and confirms the calibration for the rest of the experiment.

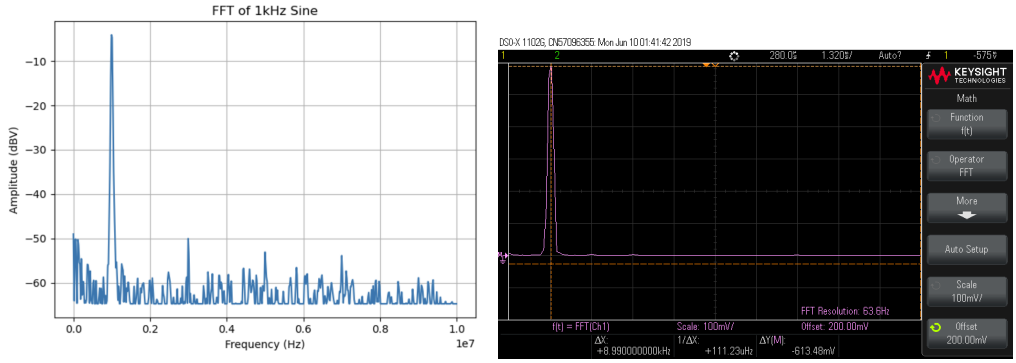


Figure 1: FFT of a 1 kHz sine wave (200 mV peak to peak)

2.Square Wave Spectrum (50% Duty Cycle)

A 50% duty cycle square wave was analysed using the FFT to examine its frequency content.

Observation: The frequency spectrum shows:

- A strong main component at the square wave's fundamental frequency.
- Clear odd harmonics at $3f$, $5f$, $7f$, and higher orders.
- Even harmonics are almost absent, as expected.
- The harmonic amplitudes decrease progressively with increasing harmonic number.

Relation to Course Content: From the Fourier series theory covered in the lectures, a 50% duty cycle square wave can be expressed as a sum of only odd numbered sine terms, with amplitudes that scale as:

$$\frac{1}{n} \qquad (n = 1, 3, 5, \ldots)$$

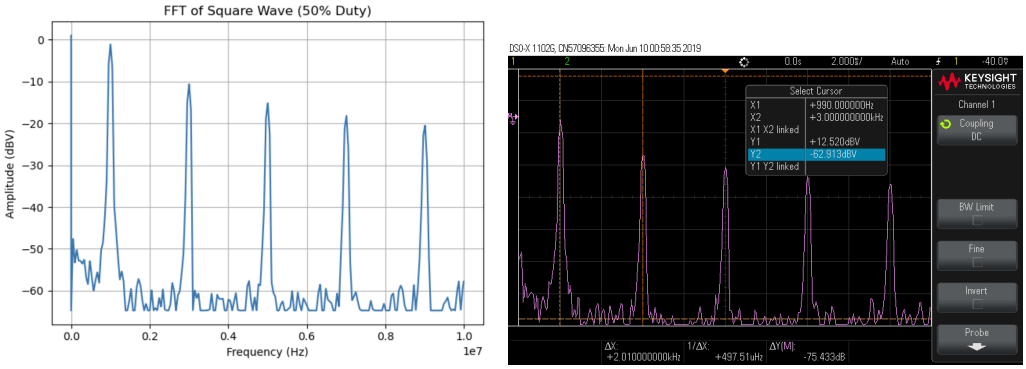


Figure 2: FFT of a 50% duty cycle square wave.

This implies:

- The square wave naturally produces only odd harmonics.
- Each successive harmonic carries less energy due to the $1/n$ amplitude decay.

The measured FFT matches this theoretical behaviour exactly, confirming the characteristic harmonic structure predicted by the Fourier series.

3.Amplitude Modulated (AM) Signal

A 1 kHz carrier was amplitude modulated using a 200 Hz message signal. FFT plots generated in Python reveal the characteristic AM spectral components.

Observation:

- One strong carrier peak at $f_c = 1\,kHz$.
- Two clear sidebands located at:

$$f_c - f_m = 800\,Hz, \qquad f_c + f_m = 1200\,Hz$$

- Sideband amplitudes are approximately $-20\,dBV$ (dependent on modulation depth).
- Changing the modulation frequency or depth alters the spacing and amplitude of the sidebands.

This produces three frequency domain components:

$$f_c, \qquad f_c - f_m, \qquad f_c + f_m$$

Thus, the measured FFT confirms the theoretical AM spectrum: a central carrier symmetrically by two sidebands. As in Fig 3.

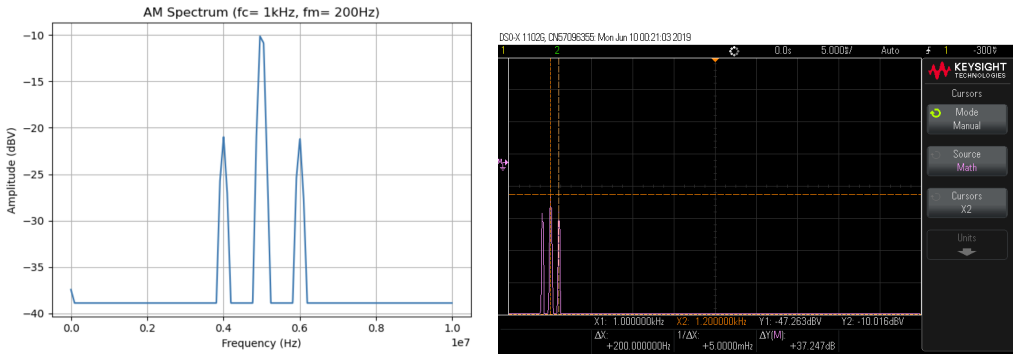


Figure 3: Amplitude Modulated (AM)

4.Frequency Modulation (FM)

The FM signal, using a 1 kHz carrier and a 200 Hz modulating frequency, was analysed for several frequency deviations ($\Delta f = 100$–500 Hz). The resulting FFT plots show the classic comb like FM spectral structure.
A larger modulation index produces:

- More sidebands
- A larger overall spectral bandwidth

The measured FFT spectra match the expected FM behaviour: increasing Δf increases the number of visible sidebands and the observed spectral bandwidth.

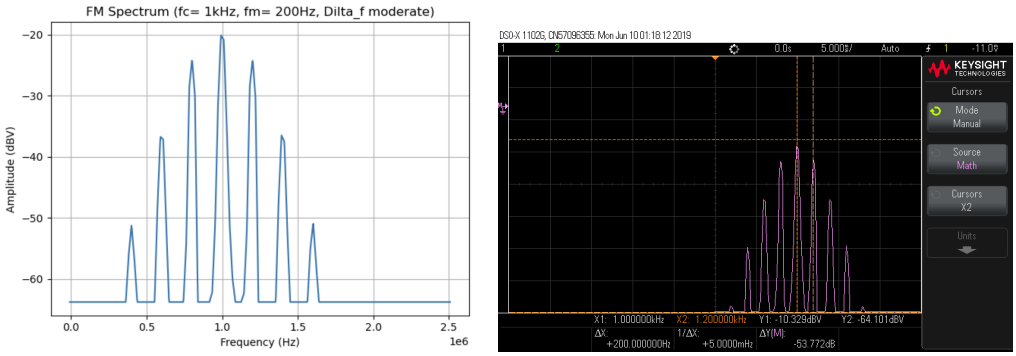


Figure 4: Frequency Modulated (FM)

Conclusion

The lab demonstrates how different signals behave in the frequency domain, reinforcing the theoretical concepts covered in the course.

1. Sine Wave: A single dominant spectral peak confirms that a pure sinusoid contains only one frequency component.

2. Square Wave: The presence of only odd harmonics verifies the Fourier series representation of nonsinusoidal periodic signals.

3. AM Signal: The carrier accompanied by two symmetric sidebands matches the analytical AM model and the expected frequency domain structure.

4. FM Signal: Multiple sidebands whose number increases with Δf confirm the behaviour predicted by the FM modulation index and Carson's Rule.

Overall, the combination of oscilloscope FFT measurements and Python analysis provides a clear, practical demonstration of Fourier transforms and modulation theory.